# 2016-2017 Virtual Challenge RQ Meet – Science Test – Chemistry Section

C17. An unknown compound contains only carbon, hydrogen, and oxygen. When a sample of mass 5.287 g is burned in excess oxygen, 12.24 g of carbon dioxide and 2.522 g of water are formed. What is the empirical formula for the compound?

A) $C_8H_8O_3$
B) $C_3H_8O_3$
C) $C_3H_3O$
D) $C_2H_2O$
E) $C_6H_5O$

C18. What is the normal boiling point for liquid bromine given its molar enthalpy and molar entropy of vaporization are 30.7 kJ/mol and 930.0 J/K-mol respectively?

A) 303 C
B) 57 C
C) 30 C
D) 330 C
E) 100 C

C19. Consider the mechanism

$2\ NO(g) \leftrightarrows N_2O_2(g)$ $k_1$ (fast)

$N_2O_2(g) + O_2(g) \rightarrow 2NO_2(g)$ $k_2$ (slow)

with the rate law: rate = $k_1k_2[NO]^2[O_2]$

A plot of potential energy vs progress of reaction would show

A) Two maxima, the second being the highest
B) Two maxima, the first being the highest
C) One maximum for the second step
D) Two minima representing the two reactions
E) One maximum for the first step

C20. How many sigma and how many pi bonds are present in the molecule $CH_2CHCN$?

A) 6 sigma and 1 pi
B) 6 sigma and 3 pi
C) 6 sigma and 2 pi
D) 5 sigma and 3 pi
E) 9 sigma and 0 pi